

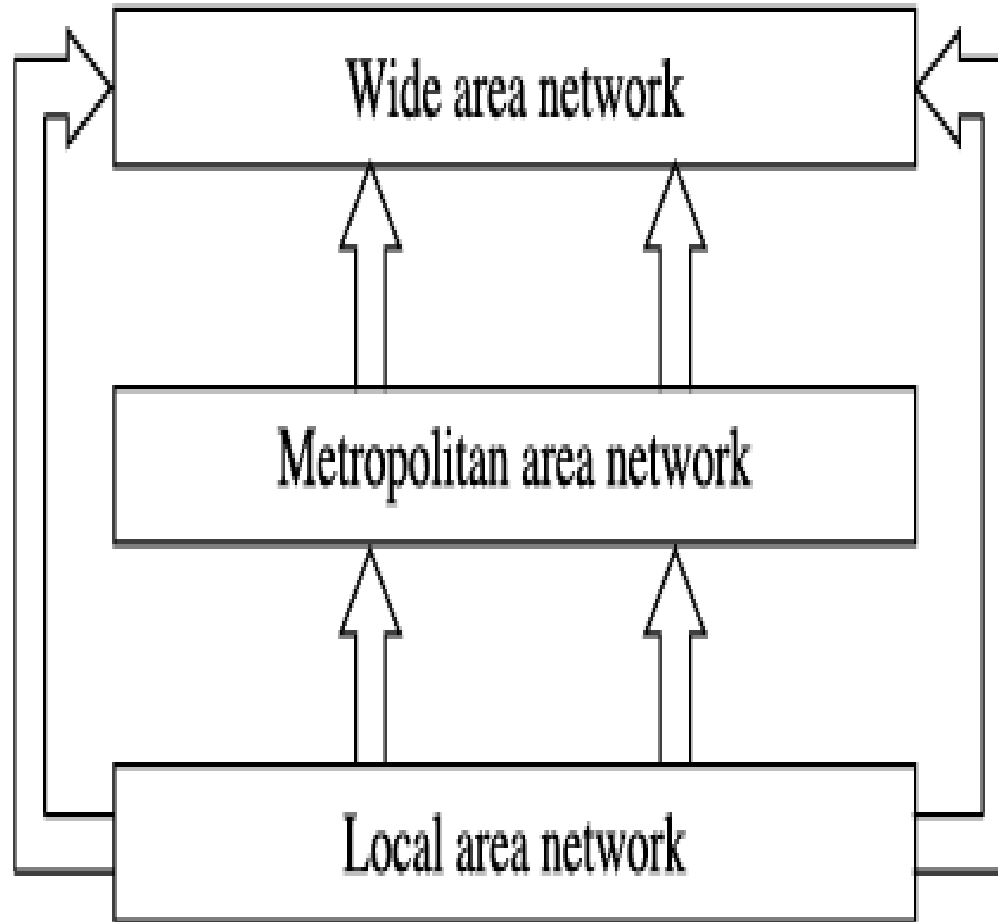
DATA NETWORKS

DATA NETWORKS

Data networks are classified according to their geographical coverage:

- Wide area networks (WANs)
- Metropolitan area networks (MANs)
- Local area networks (LANs)

DATA NETWORKS



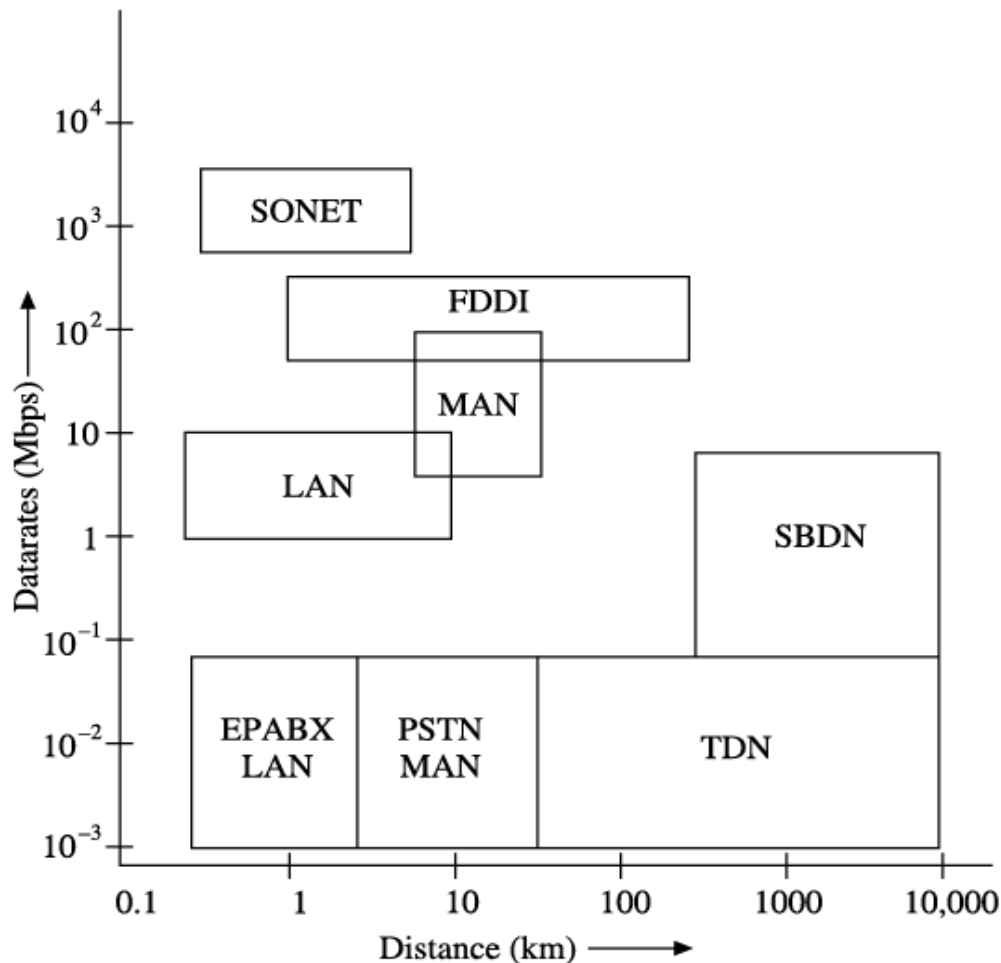
DATA NETWORKS

- LANs. MANs and WNs are generally interconnected in a hierarchical manner to form a global network
- LANs are often connected directly to WANs particularly in places where MANs are not installed or have not developed well.
- Apart from the different geographical coverage's, the range of data rates supported on these networks also differs widely.

DATA TRANSMISSION IN PSTNS

- Public switched telephone networks and electronic PABXs are designed to carry analog voice signals.
- LANs can be designed around PABXs, and MANs around PSTNs.
- In these cases, the data rates are usually limited to a maximum of 64 kbps.
- Terrestrial data networks and the integrated services digital networks however, support data rates of 1.544 or 2.048 Mbps.

DATA TRANSMISSION IN PSTNS

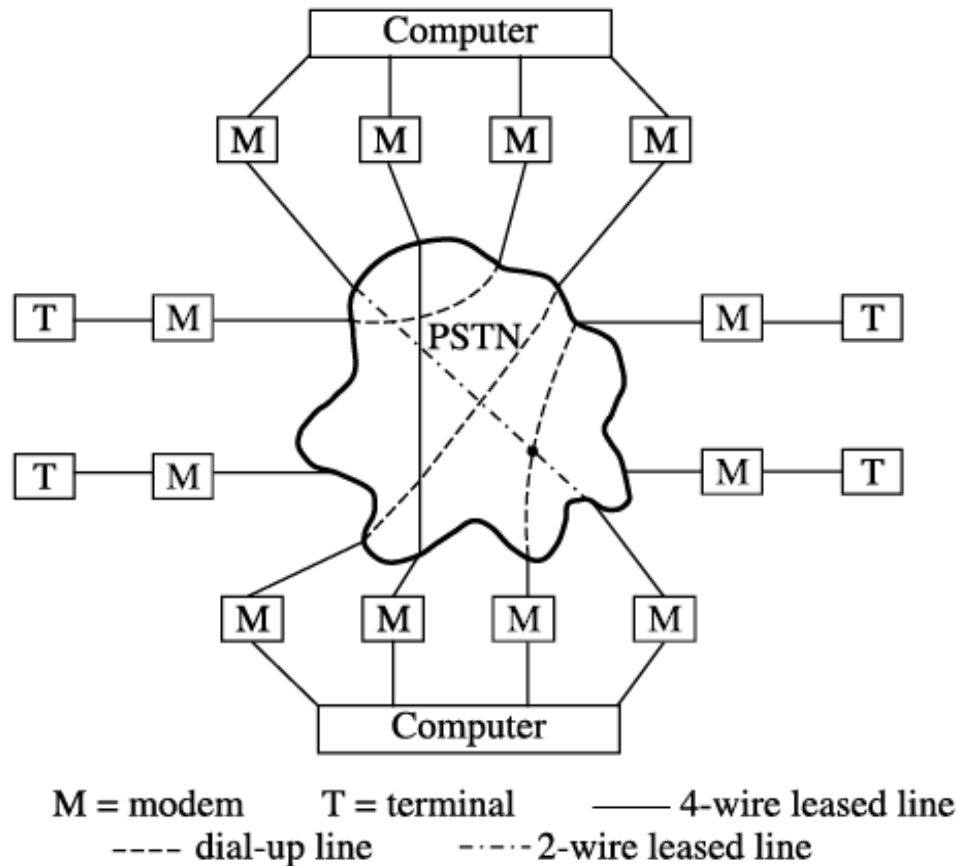


Geographical coverage and speeds of data networks

DATA TRANSMISSION IN PSTNS

- Transmission of digital data signals over PSTN networks demands that the digital signals be converted to analog form at the transmitting end and vice versa at the receiving end.
- A modulator translates the data pulses into voice band analog signals at the transmitting end.
- At the receiving end, the analog signals are demodulated to recover the digital Information.
- A combined modulator demodulator unit is called a modem.

DATA TRANSMISSION IN PSTNS



Data communication using PSTN

Data Rates in PSTNs

- A voice channel in a PSTN is band limited with a nominal bandwidth of 3.1 kHz.
- A first-cut estimate of this can be obtained from Nyquist theorem which applies to noiseless channels and states

$$R = 2H \log_2 V \text{ bps}$$

where

R = maximum data rate

H = bandwidth of the channel

V = number of discrete levels in the signal

Data Rates in PSTNs

$$R_b = H \log_2 \left(1 + \frac{S}{N} \right)$$

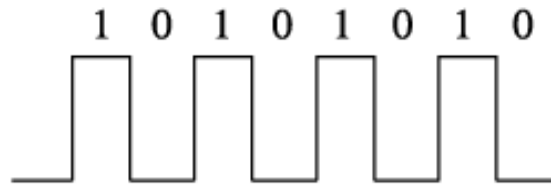
where

R_b = the maximum bit rate obtainable

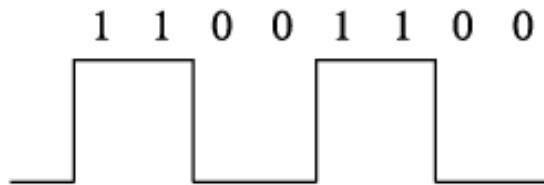
H = bandwidth of the channel

S/N = signal-to-noise ratio

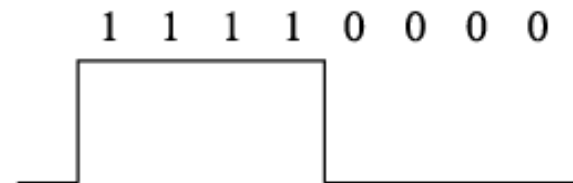
Data Rates in PSTNs



(a) $R = R_b$



(b) $R = R_b/2$



(c) $R = R_b/4$

Baud rates and bit rates

Modems

- Amplitude, frequency and phase modulation are all used in the design of modems.
- In amplitude modulation, zeros and ones are represented by two different voltage levels.
- A signal waveform $s(t)$, called baseband signal, is generated from the digital data.
- This is then multiplied by a sinusoidal carrier, say $\cos(2\pi f_c t)$, to generate a modulated signal

Modems

$$m(t) = s(t) \cos (2\pi f_0 t) \quad (10.3)$$

At the receiver end, the modulated signal is again multiplied by $\cos (2\pi f_0 t)$, yielding a received signal

$$\begin{aligned} r(t) &= s(t) \cos^2(2\pi f_0 t) \\ &= \frac{s(t)}{2} + \frac{s(t)}{2} \cos(2\pi(2f_0)t) \end{aligned} \quad (10.4)$$

Modems

- While the message sample rate is the baud rate, the bit rate is four times the baud rate in this example.
- This technique of varying the amplitude of sinusoidal carrier using the voltage levels of the baseband signal is known as amplitude shift keying (ASK) .
- If the carrier is sinusoidal and the baseband voltage level is used to vary the frequency or phase of the carrier, the modulation is known as frequency shift keying (FSK) or phase shift keying (PSK) respectively.

Switching techniques for data transmission

- They are basically designed to carry voice traffic and there are some significant differences in the nature of voice and data traffic.
- Voice traffic is generally continuous (except for the silence periods in normal speech), whereas data traffic is bursty in nature.
- In contrast, voice traffic needs low bandwidth (3.4 kHz) for long durations.

Switching techniques for data transmission

Voice traffic	Data traffic
Continuous Low bandwidth for long duration Typical line utilisation 85-95% Half duplex Real time Loss acceptable Error tolerable	Bursty High bandwidth for short duration Typical line utilisation 5-15% Half or full duplex Nonreal time or near real time Loss unacceptable Error unacceptable

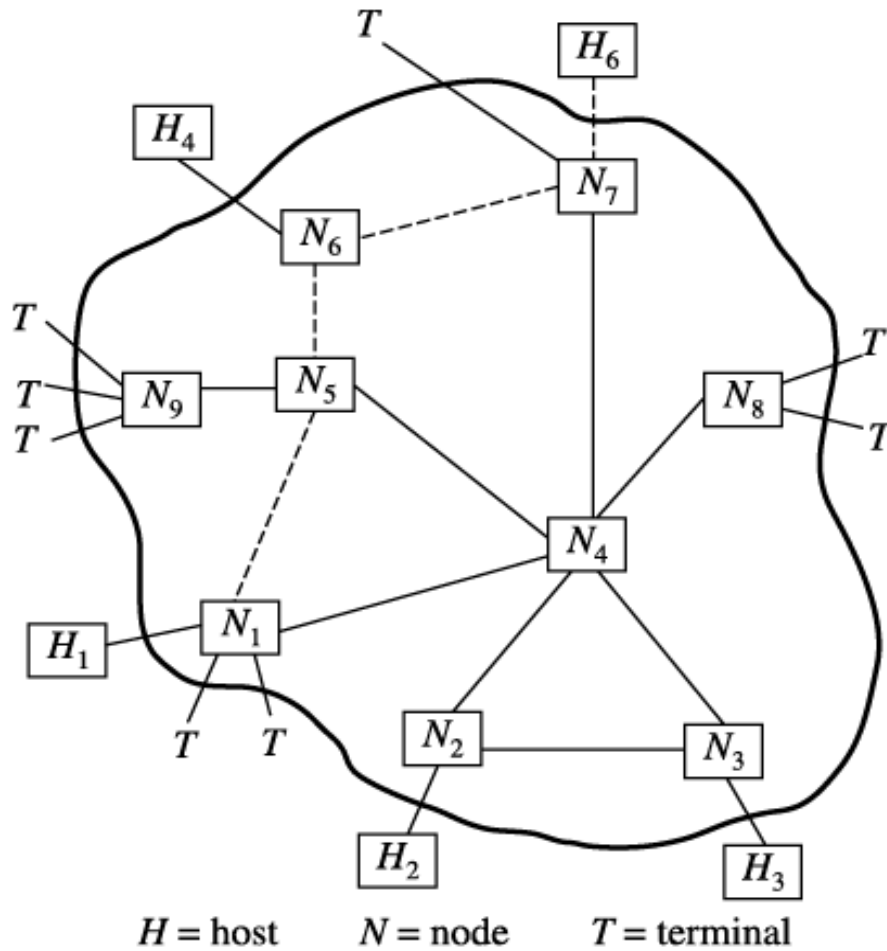
Switching techniques for data transmission

- The recognition of the diverse characteristics of voice and data traffic has led to the development of a switching technique other than the one used for voice transmission.
- This technique is better suited for transmitting data traffic.
- Hence, two switching techniques are prevalent for data transmission:
 - ❖ **Circuit switching**
 - ❖ **Store and forward (S&F) switching.**

Circuit switching

- In circuit switching, an electrical path is established between the source and the destination before any data transfer takes place.
- The electrical path may be realised by physical wires or coaxial cables or radio or satellite links.
- It remains dedicated to the communicating pair for the entire duration of the transmission irrespective of either data is actually transferred or not.
- The connection is released only when specifically signalled so by either of the communicating entities

Circuit switching



Circuit switching Network

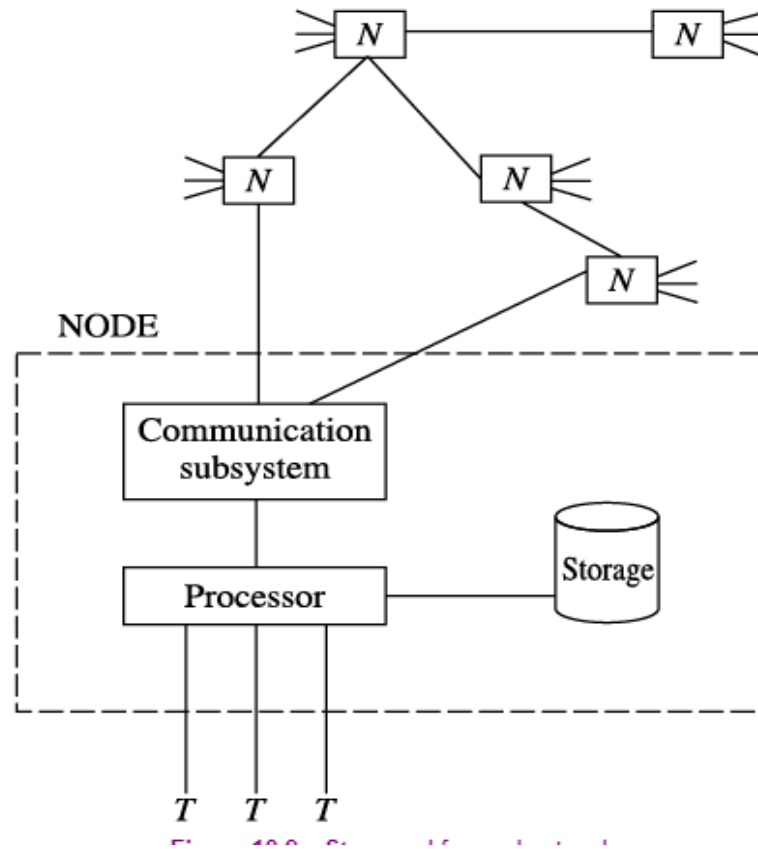
Circuit switching

- Circuit switching has certain disadvantages for transmitting data traffic.
- The path set up time which is typically of the order of 20—30 s or more turns out to be an excessive overhead for busy computer traffic which typically lasts for a few seconds or less.
- The entire line quality is affected if there is one bad link in the circuit. In fact, this is a typical problem faced in many PSTN circuit switched connections.

Store and forward switching

- A store and forward (S&F) network configuration is In S&F switching.
- The switching nodes have the ability to store user messages and forward the same towards the destination as and when the links become available.
- For this purpose, each node is equipped with a processor and some buffer storage.
- No end-to-end link is set up prior to data transmission.

Store and forward switching



Store and forward switching

- The network moves the user information from node to node.
- One such movement is called a hop.
- Since the communication links are used one at a time between any two nodes, line speeds can be utilized efficiently.
- S&F switching may be classified as:
 - Message switching
 - Packet switching

Store and forward switching

- Receive the full user message and store the same.
- Check the message for data transmission errors and perform error recover if required.
- Determine the destination address from the user message.
- Choose an appropriate link towards destination based on certain routing criterion.
- Forward the message to the next node on the chosen link.

Data communication architecture

- Data communication among computers involves a number of functions such as physical transmission of bits, error control, and routing and session establishment.
- In order to efficiently implement these functions, vendors of computer systems evolved their own architectures.
- Examples of vendor specific architectures are System Network Architecture (SNA) of IBM and Digital Network Architecture (DNA) of Digital Equipment Corporation (DEC).

Data communication architecture

An architecture which is used as a standard by all the vendors is required.

The heterogeneity covers the following aspects:

1. Systems of different vendors
2. Systems under different managements
3. Systems of different complexities
4. Systems of different technologies.

Data communication architecture

- ARPANET: The network project supported by Advanced Research Projects Agency of the Department of Defence, United States, was one of the pioneering efforts in interconnection heterogeneous systems.
- The efforts put in and the experience gained in the project significantly contributed the emergence of a world standard architecture for computer communication, largely pursued and set out by International Standardisation Organisation (ISO).

ISO-OSI reference model

System:

A system is one or more autonomous computers and their associated software, peripherals and users, which are capable of information processing and or transfer .

Subsystem:

A logically independent smaller unit of a system A succession of subsystem make up a system.

ISO-OSI reference model

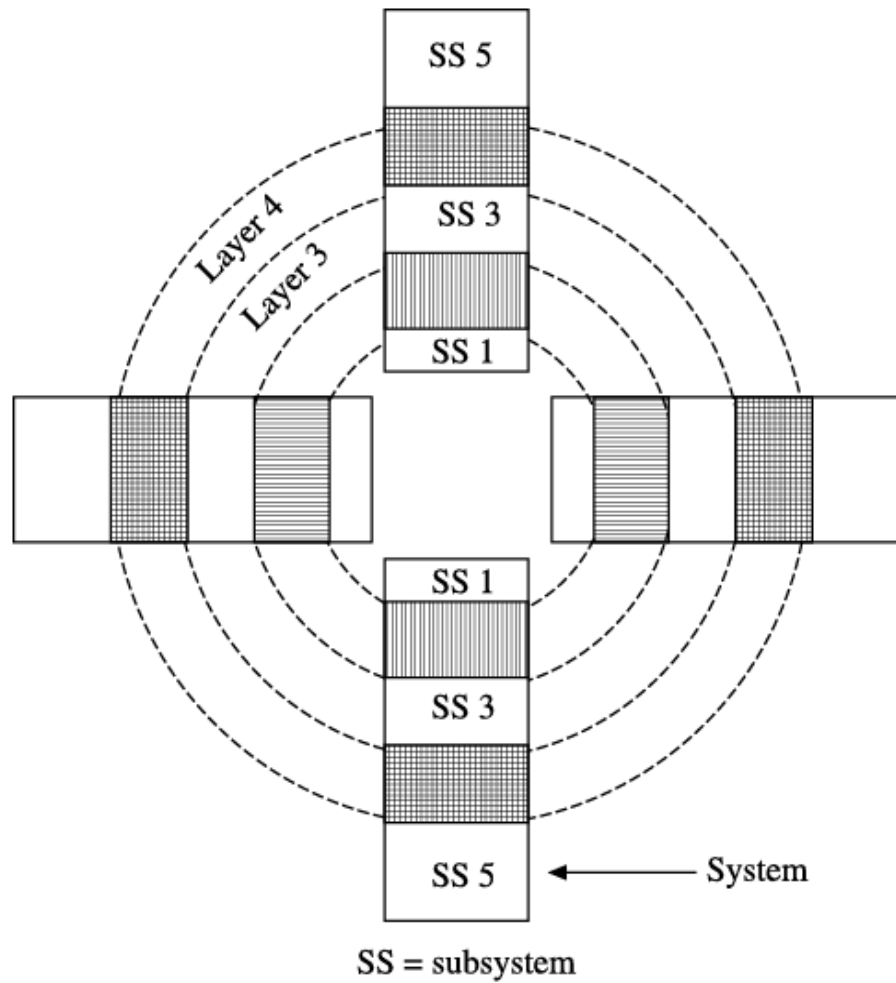
Layer:

- A layer is composed of subsystems of the same rank of all the interconnected systems.
- The subsystems and the layers are numbered starting with one at the bottom level.

Entity:

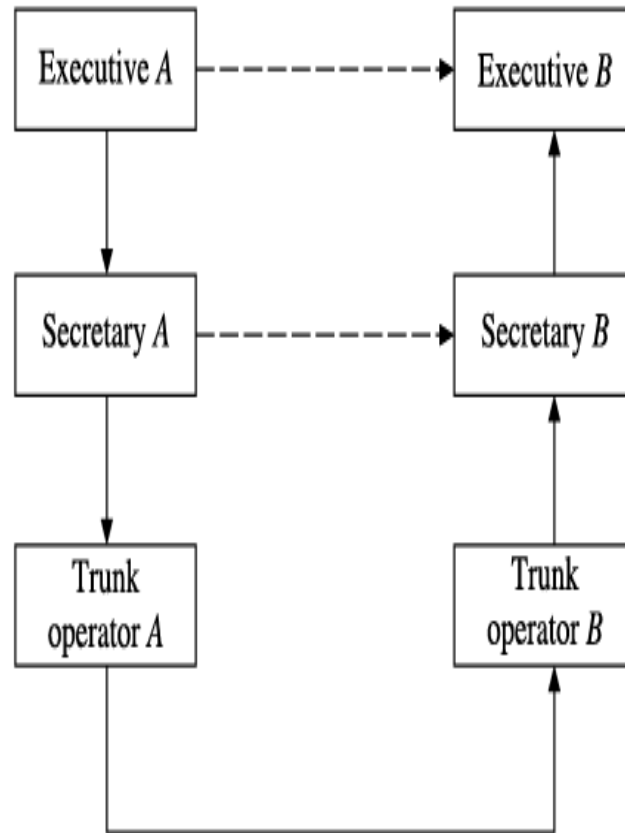
- The functions in a layer are performed by hardware subsystems and or software packages.
- These are known as entities.
- ISO-OSI architecture is a layered one.
- Layering is a natural choice for communication architectures

ISO-OSI reference model



Systems, subsystems and layers in ISO-OSI model.

ISO-OSI reference model

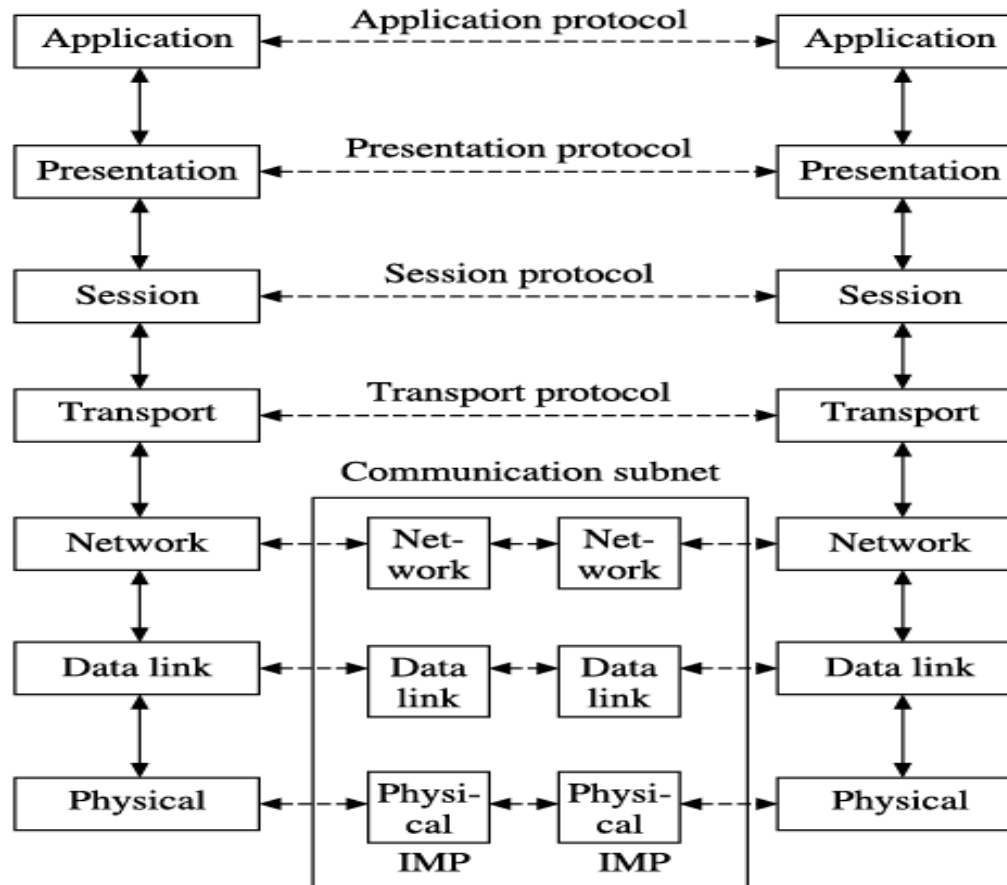


A three-layer structure for a trunk call connection.

ISO-OSI reference model

- A three-layer structure is used in this communication process.
- The conversation between an upper and lower layer is strictly business like.
- There is generally a little private conversation between the trunk operators and the two secretaries on account of their familiarity.
- A layer obtains services from its immediate lower layer and provides services to its immediate upper layer.
- There are fairly well defined functions to be performed by each layer.
- It is immaterial as to how the functions of each layer are implemented.

ISO-OSI reference model



ISO-OSI reference model

The important principles are:

- Create layers to handle Functions which are manifestly different in the process performed or technology involved.
- Collect similar functions into the same layer and create a boundary at a point where the number of interactions across the boundary are minimised.
- Create a layer of easily localised functions so that the layer could be totally redesigned and its protocols changed in a major way to take advantage of new advances in architectures.

Link to link layers

- ✓ The first three layers, viz. physical, data link and network layers, form the link-to-link layers of OSI reference model.
- ✓ They obtain services from the immediate lower layer and provide services to the immediate upper layer.
- ✓ OSI services may be placed under two broad categories:
 - Connection oriented services
 - Connectionless services.

Link to link layers

Connection oriented services

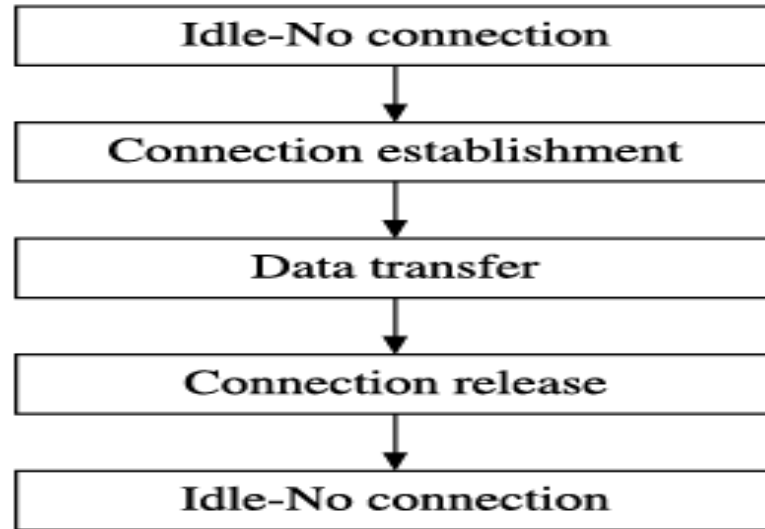
- A connection is first established between the sender and the receiver before data transfer can commence.
- The connection may be virtual (logical) or physical, depending upon the network capabilities and facilities.
- The essence of a connection oriented service is that a connection acts like a tube or a pipe delivering.
- The data to the receiver strictly in the same order in which the data was put into the connection by the sender. Connection oriented service is modelled after the telephone system.

Link to link layers

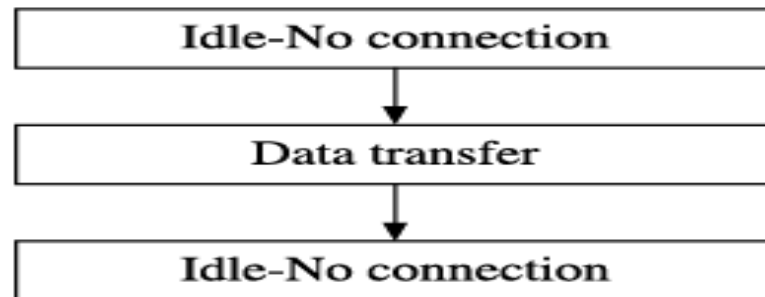
Connectionless service

- It is modelled after the postal system.
- Each submission by the sender is treat independently of others and is sell-contained with the full address of the destination and the source indication which may be the full address too.
- it is possible that the fast one is delayed and the second one arrives first. Datagram service and the virtual circuit service

Link to link layers



(a) Connection oriented service



(b) Connectionless service

Physical layer

- It is essential that the OSI architecture permits the usage of a realistic variety of physical media and control procedures.
- Keeping this in mind, the lowest layer of the architecture has been identified as the physical layer
- This layer performs functions associated with the activation and deactivation of physical connections.

Data link layer

- Special techniques are required to ensure error free transmission of data.
- The data link layer deals with error detection and automatic recover procedures required when a message is lost or corrupted.
- For this purpose, a user of this layer, ie. the network layer, is required to break up the data to be transmitted into frames which are then numbered and transmitted sequentially.

Data link layer

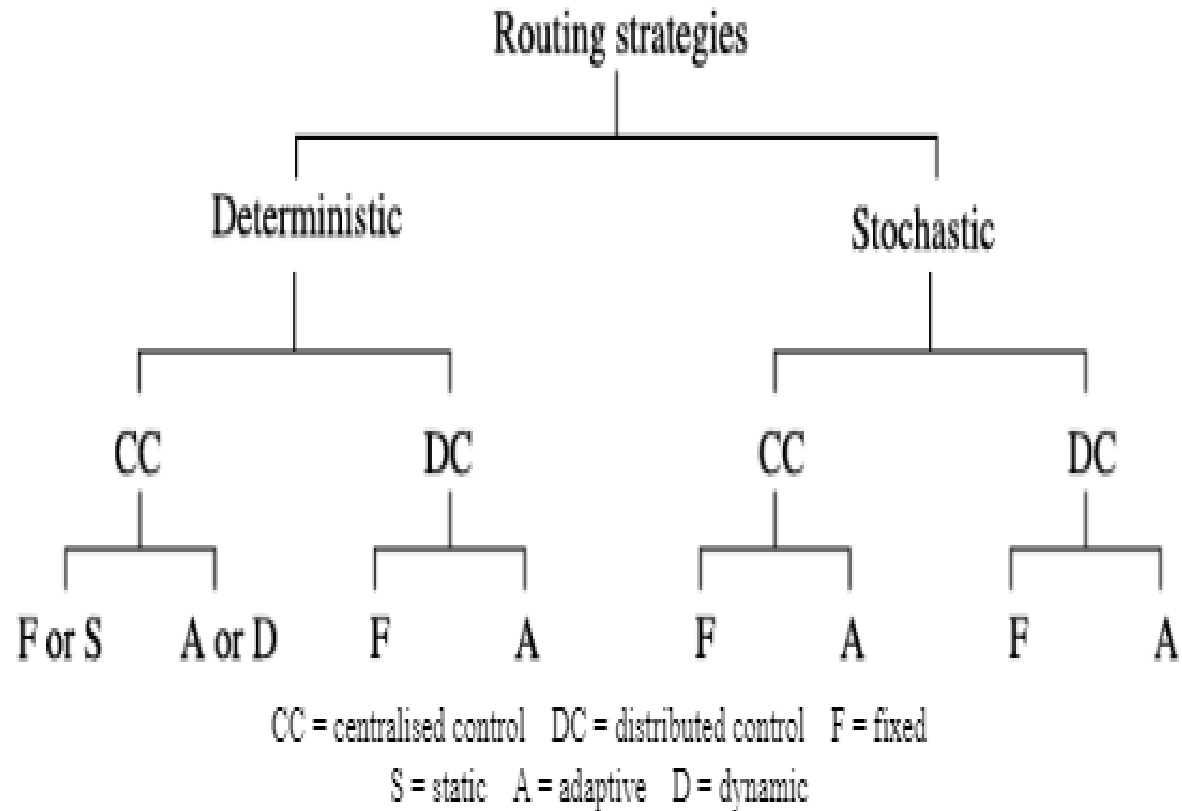
There are three error control mechanisms that are commonly used:

1. Echo checking
2. Forward error correction (FEC)
3. Automatic repeat request (ARQ).

Network Layer

- The highest link-to-link layer in the OSI model is the network layer.
- Although this layer functions on a link-to-link basis, it is concerned with transmission of packets from the source node to the destination node.
- The network layer makes invisible to the transport layer, the details of the underlying communication media and the different characteristics of the transmission and network technologies.

Network Layer



Network Layer

A number of measures may be used in assessing the performance of a routing algorithm:

1. Minimum delay
2. Minimum number of intermediate nodes or hops
3. Processing complexity
- 4 Signalling capacity required on the network
5. The rate of adaption in the case of adaptive algorithms
6. Fairness to all types of traffic
7. Robustness: the ability to reach the destination even when parts of the network fail

END-TO-END LAYERS

- Many data networks were operational before the OSI model was designed.
- These networks were well thought out up to the network layer, but little had been done about the transport layer and above.
- As a consequence, the design of the bottom three layers of OSI was highly influenced by pre-OSI developments.

Transport Layer

- Transport layer is the first end-to-end layer in the OSI architecture.
- It is responsible for matching user message characteristics and service requirements with that of the network capabilities.
- In a packet switched network, the transport entity breaks up a long user message into packets to match the network capabilities.
- The packets are reassembled at the destination transport subsystem to reconstruct the user message.

Transport Layer

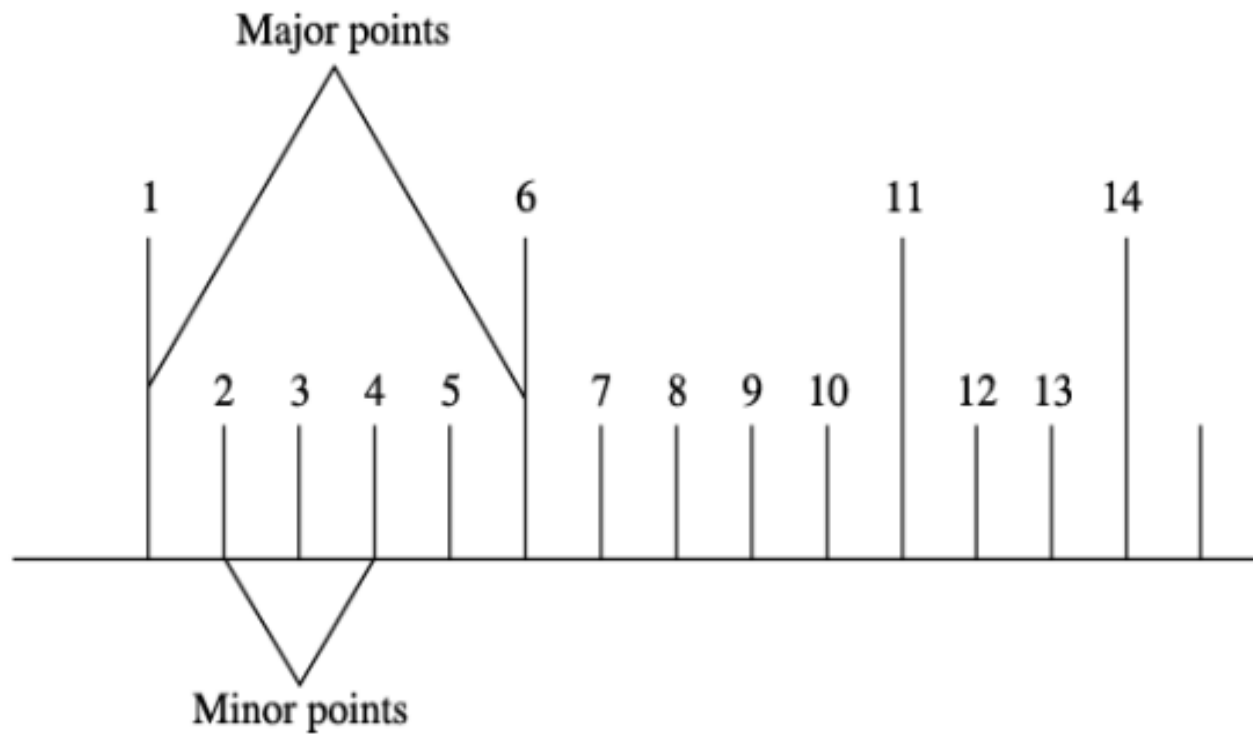
Some QOS parameters that are of direct interest to the users are:

1. Transit delay
2. Residual error rate
3. Protection
4. Transfer failure probability
5. Priority
6. Throughput.

Session Layer

- The main function of the session layer is to organise different sessions between cooperating entities and perform all related functions like synchronisation, failure management, control, etc. for the successful execution of a session.
- Online search of databases, remote job entry, remote login to a time sharing system and file transfer between two systems are all examples of sessions.
- The session layer may also offer a director's service.

Session Layer



Presentation layer

- The purpose of the presentation layer is to represent information to the communicating application entities in a way that preserves the meaning.
- Syntax differences are resolved by encoding application data into a standard abstract notation that is valid throughout the network.

SATELLITE BASED DATA NETWORKS

- Satellite network topology and configurations modulation schemes and bandwidth utilisation; these are aspects related to the physical layer functions of the reference model.
- Being a common communication resource accessible by all or a group of earth stations simultaneously. media access becomes a nontrivial function in the data link layer.
- Satellite communication being broadcast in nature, routing becomes a trivial function;

LAN

A local area network (LAN) typifies a distributed environment and finds applications in a number of areas.

Some examples are:

1. Office automation
2. Factory automation
3. Distributed computing
4. Fire and security systems
5. Process control
6. Document distribution.

LAN

The advantages offered by the LANs are:

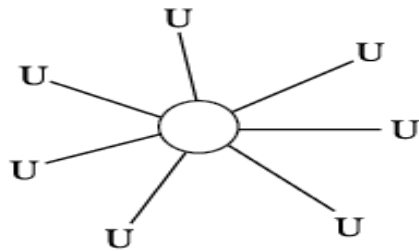
- Unlike a large centralised system, a LAN may evolve with time. It may be put into operation with a small investment, and more systems may be added as the need arises.
- Since LAN is a set of multiple interconnected systems, it offers a good back up capability in the event of one or two systems failing in the network. This, in turn, enhances the reliability and availability of the systems to users.

LAN

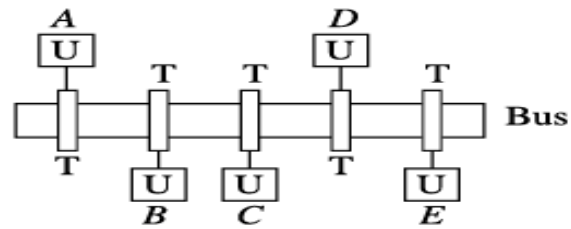
The advantages offered by the LANs are:

- LAN provides a resource-sharing environment. Expensive peripherals, hosts and databases may be shared by all the LAN users.
- A LAN adhering to a certain standard permits multivendor systems to be connected to it. Thus, a user is not committed to a single vendor.
- In LAN, the systems are generally so chosen as to meet most of the user

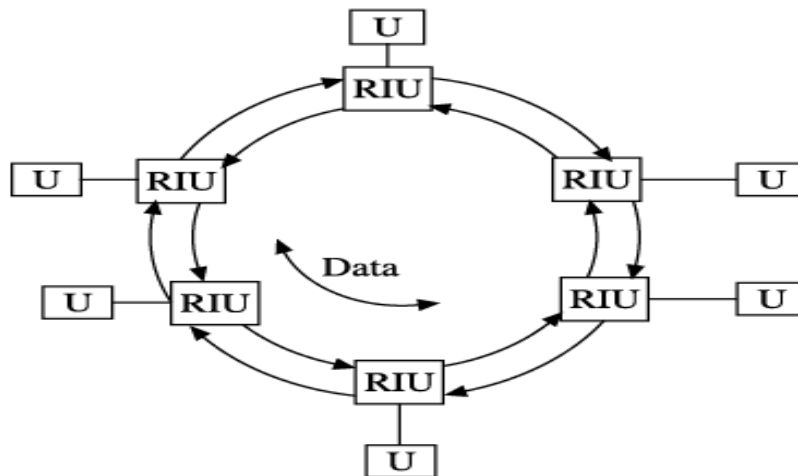
LAN



(a) Star topology



(b) Bus topology



RIU = ring interface unit

(c) Ring topology

As a result, only three combinations of access techniques and topologies are popularly used:

1. Multiple access bus
2. Token passing ring or token ring
3. Token passing bus or token bus.

Metropolitan area network

- A metropolitan area network (MAN) usually covers a geographical area spanning a distance of 5—50 km.
- In functionality MANs support services that require guaranteed bandwidth and bounded delay performance
- In addition to data services that do not pose such restrictions.
- Additional services may include voice and video: depending on the available bandwidth.

Metropolitan area network

- MANs may operate at speeds of 1 Mbps and above although a more common range is 50—150 kbps.
- Network structures to MAN are similar to the ones used in LANs: star, bus and ring. In this sense.
- The LANs may be considered as an extension of LANs

Fiber optic networks

Optical fibre networks are characterised by

- High speed operation (typically 100 Mbps or more)
- Ability to span large distances (100—200 km)
- Ability to support a moderate number of stations; typically 10 to a few hundred stations are supported with a maximum limit around 1000.

Fiber optic networks

- These characteristics make the fibre optic networks suitable for high speed LAs and MANs with a limited number of stations.
- Fibre networks may be configured around a star, ring or bus structure.
- The number of stations that can be supported in a star or a bus structure is relatively low compared to that in a ring configuration.
- Optical fibres are inherently unidirectional and this influences the way in which the network structures are realised.

Fiber optic networks

- The passive hub is usually a silica cylinder. The incoming fibres are fused to one end of the cylinder and the outgoing fibres to the other end.
- The light emitted by the transmitters (LEDs) of the stations diffuses inside the passive hub illuminating all the receivers (,photodiodes).
- The incoming energy is divided among all the outgoing lines.